

```
1 #include <stdio.h>
2 int main(){
3     int a[3][3]={3,6,7},{4,6,9},{2,5,8}};
4     int b[3][3]={2,5,-9},{5,4,6},{8,6,7}};
5     int c[3][3],i,j;
6     for(i=0;i<3;i++)
7     {
8         for(j=0;j<3;j++)
9         {
10             c[i][j]=a[i][j]+b[i][j];
11             printf("%d ",c[i][j]);
12         }
13         printf("\n");
14     }
15     return 0;
16 }
17
```

```
1 #include <stdio.h>
2 int main(){
3     int a[100],n,i,ele;
4     scanf("%d",&n);
5     for(i=0;i<n;i++)
6         scanf("%d",&a[i]);
7     scanf("%d",&ele);
8     for(i=n-1;i>=0 && a[i]>ele;i--)
9         a[i+1]=a[i];
10    a[i+1]=ele;
11    n++;
12    for(i=0;i<n;i++)
13        printf("%d ",a[i]);
14    return 0;
15 }
```

```
1 #include <stdio.h>
2 int main(){
3     int a[100],n,i,j,key;
4     scanf("%d",&n);
5     for(i=0;i<n;i++)
6         scanf("%d",&a[i]);
7     scanf("%d",&key);
8     for(i=0;i<n;i++)
9         if(a[i]==key){
10             for(j=i;j<n-1;j++)
11                 a[j]=a[j+1];
12             n--;
13             i--;
14
15         }
16         for(i=0;i<n;i++)
17             printf("%d ",a[i]);
18         return 0;
19 }
```

```
1 #include <stdio.h>
2 int main(){
3     int a[100],n,i,flag=1;
4     scanf("%d",&n);
5     for(i=0;i<n;i++)
6         scanf("%d",&a[i]);
7     for(i=0;i<n/2;i++){
8         if(a[i]!=a[n-1-i]){
9             flag=0;
10            break;
11        }
12    }
13    if(flag){
14        printf("Palindrome");
15    }
16    else
17        printf("Not palindrome");
18    return 0;
19 }
20
```

```
1 #include <stdio.h>
2 int main(){
3     int a[100],b[100],n1,n2,i,j;
4     scanf("%d",&n1);
5     for(i=0;i<n1;i++)
6         scanf("%d",&a[i]);
7     scanf("%d",&n2);
8     for(i=0;i<n2;i++)
9         scanf("%d",&b[i]);
10    for(i=0;i<n1;i++)
11        for(j=0;j<n2;j++)
12            if(a[i]==b[j])
13                printf("%d",a[i]);
14    return 0;
15 }
```

```
1 #include <stdio.h>
2 int main()
3 {
4     int a[3][3]={{3,6,7},{4,6,9},{2,5,8}};
5     int i,sum=0;
6     for(i=0;i<3;i++)
7         sum+=a[i][i];
8     printf("sum of diagonal =%d",sum);
9     return 0;
10 }
```

```
1 #include <stdio.h>
2 int main(){
3     int a[3][3],b[3][3],c[3][3];
4     int i,j,k;
5     for(i=0;i<3;i++)
6         for(j=0;j<3;j++)
7             scanf("%d",&a[i][j]);
8     for(i=0;i<3;i++)
9         for(j=0;j<3;j++)
10            scanf("%d",&b[i][j]);
11     for(i=0;i<3;i++){
12         for(j=0;j<3;j++){
13             c[i][j]=0;
14             for(k=0;k<3;k++)
15                 c[i][j]+=a[i][k]*b[j][k];
16         }
17     }
18     for(i=0;i<3;i++){
19         for(j=0;j<3;j++)
20             printf("%d ",c[i][j]);
21     printf("\n");
```



```
1 #include <stdio.h>
2
3 int main() {
4     int r, c, i, j;
5     int matrix[10][10], transpose[10][10];
6
7     printf("Enter rows and columns: ");
8     scanf("%d %d", &r, &c);
9
10    printf("\nEnter matrix elements:\n");
11    for (i = 0; i < r; ++i) {
12        for (j = 0; j < c; ++j) {
13            printf("Enter element a%d%d: ", i + 1, j + 1);
14            scanf("%d", &matrix[i][j]);
15        }
16    }
17
18
19    for (i = 0; i < r; ++i) {
20        for (j = 0; j < c; ++j) {
21            transpose[j][i] = matrix[i][j];
```



```
1 #include <stdio.h>
2
3 int main() {
4     int rows, cols, i, j;
5     int matrix[100][100];
6     int sum = 0;
7
8     printf("Enter the number of rows (max 100): ");
9     if (scanf("%d", &rows) != 1) return 1;
10
11     printf("Enter the number of columns: ");
12     if (scanf("%d", &cols) != 1) return 1;
13
14     printf("Enter the elements of the matrix:\n");
15     for (i = 0; i < rows; i++) {
16         for (j = 0; j < cols; j++) {
17             scanf("%d", &matrix[i][j]);
18         }
19     }
20
21     printf("The matrix is:\n");
```

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, j;
5     int matrix[10][10];
6     int product = 1;
7
8     printf("Enter the size of matrix: ");
9     scanf("%d", &n);
10
11     printf("Enter matrix elements:\n");
12     for (i = 0; i < n; i++) {
13         for (j = 0; j < n; j++) {
14             scanf("%d", &matrix[i][j]);
15         }
16     }
17
18     for (i = 0; i < n; i++) {
19         product *= matrix[i][i];
20     }
21 }
```

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, j, sum = 0;
5     int matrix[10][10];
6
7     printf("Enter the size of matrix: ");
8     scanf("%d", &n);
9
10    printf("Enter matrix elements:\n");
11    for (i = 0; i < n; i++) {
12        for (j = 0; j < n; j++) {
13            scanf("%d", &matrix[i][j]);
14        }
15    }
16
17    for (i = 0; i < n; i++) {
18        for (j = i; j < n; j++) {
19            sum += matrix[i][j];
20        }
21    }
```

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, j, sum = 0;
5     int matrix[10][10];
6
7     printf("Enter the size of matrix: ");
8     scanf("%d", &n);
9
10    printf("Enter matrix elements:\n");
11    for (i = 0; i < n; i++) {
12        for (j = 0; j < n; j++) {
13            scanf("%d", &matrix[i][j]);
14        }
15    }
16
17    for (i = 0; i < n; i++) {
18        for (j = 0; j <= i; j++) {
19            sum += matrix[i][j];
20        }
21    }
```

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, j, sum = 0;
5     int matrix[10][10];
6
7     printf("Enter the size of matrix: ");
8     scanf("%d", &n);
9
10    printf("Enter matrix elements:\n");
11    for (i = 0; i < n; i++) {
12        for (j = 0; j < n; j++) {
13            scanf("%d", &matrix[i][j]);
14        }
15    }
16
17    // Sum of left diagonal elements
18    for (i = 0; i < n; i++) {
19        sum += matrix[i][i];
20    }
21
```

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, j, sum = 0;
5     int matrix[10][10];
6
7     printf("Enter the size of matrix: ");
8     scanf("%d", &n);
9
10    printf("Enter matrix elements:\n");
11    for (i = 0; i < n; i++) {
12        for (j = 0; j < n; j++) {
13            scanf("%d", &matrix[i][j]);
14        }
15    }
16
17    for (i = 0; i < n; i++) {
18        sum += matrix[i][n - 1 - i];
19    }
20
21    printf("Sum of right diagonal elements = %d\n", sum);
```



```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, j, symmetric = 1;
5     int matrix[10][10];
6
7     printf("Enter the size of matrix: ");
8     scanf("%d", &n);
9
10    printf("Enter matrix elements:\n");
11    for (i = 0; i < n; i++) {
12        for (j = 0; j < n; j++) {
13            scanf("%d", &matrix[i][j]);
14        }
15    }
16
17    for (i = 0; i < n; i++) {
18        for (j = 0; j < n; j++) {
19            if (matrix[i][j] != matrix[j][i]) {
20                symmetric = 0;
21                break;
```



```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, j, diagonal = 1;
5     int matrix[10][10];
6
7     printf("Enter the size of matrix: ");
8     scanf("%d", &n);
9
10    printf("Enter matrix elements:\n");
11    for (i = 0; i < n; i++) {
12        for (j = 0; j < n; j++) {
13            scanf("%d", &matrix[i][j]);
14        }
15    }
16
17    for (i = 0; i < n; i++) {
18        for (j = 0; j < n; j++) {
19            if (i != j && matrix[i][j] != 0) {
20                diagonal = 0;
21                break;
```

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, j, upper = 1;
5     int matrix[10][10];
6
7     printf("Enter the size of matrix: ");
8     scanf("%d", &n);
9
10    printf("Enter matrix elements:\n");
11    for (i = 0; i < n; i++) {
12        for (j = 0; j < n; j++) {
13            scanf("%d", &matrix[i][j]);
14        }
15    }
16
17    for (i = 1; i < n; i++) {
18        for (j = 0; j < i; j++) {
19            if (matrix[i][j] != 0) {
20                upper = 0;
21                break;
```

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, j, lower = 1;
5     int matrix[10][10];
6
7     printf("Enter the size of matrix: ");
8     scanf("%d", &n);
9
10    printf("Enter matrix elements:\n");
11    for (i = 0; i < n; i++) {
12        for (j = 0; j < n; j++) {
13            scanf("%d", &matrix[i][j]);
14        }
15    }
16
17    for (i = 0; i < n; i++) {
18        for (j = i + 1; j < n; j++) {
19            if (matrix[i][j] != 0) {
20                lower = 0;
21                break;
```

```
1 #include <stdio.h>
2
3 int main() {
4     int r, c, i, j, equal = 1;
5     int a[10][10], b[10][10];
6
7     printf("Enter rows and columns: ");
8     scanf("%d %d", &r, &c);
9
10    printf("Enter first matrix:\n");
11    for (i = 0; i < r; i++) {
12        for (j = 0; j < c; j++) {
13            scanf("%d", &a[i][j]);
14        }
15    }
16
17    printf("Enter second matrix:\n");
18    for (i = 0; i < r; i++) {
19        for (j = 0; j < c; j++) {
20            scanf("%d", &b[i][j]);
21        }
```

```
1 #include <stdio.h>
2
3 int main() {
4     int r, c, i, j;
5     int a[10][10], b[10][10], result[10][10];
6
7     printf("Enter rows and columns: ");
8     scanf("%d %d", &r, &c);
9
10    printf("Enter first matrix:\n");
11    for (i = 0; i < r; i++) {
12        for (j = 0; j < c; j++) {
13            scanf("%d", &a[i][j]);
14        }
15    }
16
17    printf("Enter second matrix:\n");
18    for (i = 0; i < r; i++) {
19        for (j = 0; j < c; j++) {
20            scanf("%d", &b[i][j]);
21        }
```

```
1 #include <stdio.h>
2
3 int main() {
4     int r, c, scalar, i, j;
5     int matrix[10][10];
6
7     printf("Enter rows and columns: ");
8     scanf("%d %d", &r, &c);
9
10    printf("Enter matrix elements:\n");
11    for (i = 0; i < r; i++) {
12        for (j = 0; j < c; j++) {
13            scanf("%d", &matrix[i][j]);
14        }
15    }
16
17    printf("Enter scalar: ");
18    scanf("%d", &scalar);
19
20    for (i = 0; i < r; i++) {
21        for (j = 0; j < c; j++) {
```



```
1 #include <stdio.h>
2
3 int main() {
4     int a[2][2], determinant;
5
6     printf("Enter 2x2 matrix elements:\n");
7
8     for (int i = 0; i < 2; i++) {
9         for (int j = 0; j < 2; j++) {
10             scanf("%d", &a[i][j]);
11         }
12     }
13
14     determinant = (a[0][0] * a[1][1]) - (a[0][1] * a[1][0]);
15
16     printf("Determinant = %d\n", determinant);
17
18     return 0;
19 }
```



```
1 #include <stdio.h>
2
3 int main() {
4     float a, b, c, d;
5     float det;
6
7     printf("Enter elements of 2x2 matrix:\n");
8     scanf("%f %f %f %f", &a, &b, &c, &d);
9
10    det = a * d - b * c;
11
12    if (det == 0) {
13        printf("Inverse does not exist.\n");
14    } else {
15        printf("Inverse of the matrix:\n");
16        printf("%.2f %.2f\n", d / det, -b / det);
17        printf("%.2f %.2f\n", -c / det, a / det);
18    }
19
20    return 0;
21 }
```

```
1 #include <stdio.h>
2
3 int main() {
4     int a[10][10], m, n, i, j, k, rank = 0;
5     float ratio;
6
7     printf("Enter rows and columns: ");
8     scanf("%d %d", &m, &n);
9
10    printf("Enter matrix elements:\n");
11    for (i = 0; i < m; i++)
12        for (j = 0; j < n; j++)
13            scanf("%d", &a[i][j]);
14
15    for (j = 0; j < n && rank < m; j++) {
16        for (i = rank; i < m; i++) {
17            if (a[i][j] != 0) {
18                for (k = 0; k < n; k++) {
19                    int temp = a[rank][k];
20                    a[rank][k] = a[i][k];
21                    a[i][k] = temp;
```

```
1 #include <stdio.h>
2
3 int main() {
4     int a[10][10], r, c, i, j, k;
5     float ratio;
6
7     printf("Enter rows and columns: ");
8     scanf("%d %d", &r, &c);
9
10    printf("Enter matrix elements:\n");
11    for (i = 0; i < r; i++)
12        for (j = 0; j < c; j++)
13            scanf("%d", &a[i][j]);
14
15    for (i = 0; i < r; i++) {
16
17        if (a[i][i] == 0) {
18            for (k = i + 1; k < r; k++) {
19                if (a[k][i] != 0) {
20                    for (j = 0; j < c; j++) {
21                        int temp = a[i][i];
```